

Fifth Assignment: Heuristics, Metaheuristics, and Approximation Algorithms

Instructions

- Answer conceptual questions clearly and precisely.
- Show all reasoning in computational parts.
- For algorithmic parts, provide:
 - A flowchart
 - Clear pseudocode

1 Approximation Algorithms

Approximation algorithms provide solutions that are guaranteed to be close to optimal.

These algorithms are particularly important for NP-hard combinatorial optimization problems.

Conceptual Questions

1. What is an approximation algorithm?
2. What is an approximation ratio? Explain both minimization and maximization cases.
3. What does it mean for an algorithm to have a worst-case guarantee?
4. What is a Polynomial-Time Approximation Scheme (PTAS)?
5. What is the difference between PTAS and FPTAS?
6. Why are approximation algorithms important in combinatorial optimization?

Techniques in Approximation Algorithms

1. Explain the following approximation techniques in detail:
 - Greedy algorithms
 - Local search
 - LP relaxation and rounding
 - Primal-dual method
2. For each technique, discuss:
 - The main idea
 - When it is applicable
 - Advantages and limitations

Exercise 1

Vertex Cover Approximation

Consider the Vertex Cover problem on an undirected graph.

1. Describe the standard 2-approximation greedy algorithm.
2. Apply the algorithm to the following graph:
Edges:

$$E = \{(1, 2), (1, 3), (2, 3), (2, 4), (3, 5)\}$$

3. Show each step of the algorithm.
4. Compute the size of the obtained solution.
5. Provide a lower bound on the optimal solution.
6. Verify the approximation ratio.

Exercise 2

Approximation Algorithms for the Traveling Salesman Problem (TSP)

Consider the metric TSP, where the distance matrix satisfies the triangle inequality.

1. What is the triangle inequality? Why is it important for approximation algorithms for TSP?
2. Describe the following approximation algorithms:
 - Double Tree (MST-based) algorithm
 - Christofides' algorithm

For each algorithm, explain:

- The main idea
 - The steps of the algorithm
 - The approximation ratio
3. Explain why the Double Tree algorithm is a 2-approximation.
 4. Explain why Christofides' algorithm achieves a $\frac{3}{2}$ -approximation ratio.

Numerical Example

Consider the following complete graph with distances:

$$\begin{pmatrix} 0 & 2 & 3 & 6 & 8 \\ 2 & 0 & 4 & 5 & 7 \\ 3 & 4 & 0 & 3 & 6 \\ 6 & 5 & 3 & 0 & 2 \\ 8 & 7 & 6 & 2 & 0 \end{pmatrix}$$

1. Verify that the triangle inequality holds.
2. Construct a Minimum Spanning Tree (MST).
3. Apply the Double Tree algorithm:
 - Double the edges of the MST
 - Find an Eulerian tour
 - Shortcut to obtain a Hamiltonian cycle
4. Compute the total tour length.

2 Heuristics

Heuristics are problem-solving techniques designed to produce good solutions quickly.

Conceptual Questions

1. What is a heuristic algorithm?
2. What is the difference between exact algorithms and heuristics?
3. What are constructive heuristics?
4. What are search-based (improvement) heuristics?
5. What are the advantages and disadvantages of heuristics?
6. In which types of problems are heuristics particularly useful?

Constructive Heuristics

1. Explain the general idea of constructive heuristics.
2. Describe at least two examples, such as:
 - Greedy construction
 - Nearest neighbor heuristic for TSP
3. Discuss how solution quality depends on construction order.

Search-Based Heuristics

1. Explain the concept of local search.
2. What is a neighborhood structure?
3. What is a local optimum?
4. What are common strategies to escape local optima?

Exercise

Nearest Neighbor Heuristic for TSP

Consider the following distance matrix:

$$\begin{pmatrix} 0 & 2 & 9 & 10 \\ 2 & 0 & 6 & 4 \\ 9 & 6 & 0 & 3 \\ 10 & 4 & 3 & 0 \end{pmatrix}$$

1. Apply the nearest neighbor heuristic starting from node 1.
2. Show each step of the construction.
3. Compute the total tour length.
4. Is the solution necessarily optimal? Explain.

3 Metaheuristics

Metaheuristics provide general frameworks for exploring solution spaces efficiently.

Conceptual Questions

1. What is a metaheuristic?
2. How do metaheuristics differ from simple heuristics?
3. What are the key components of a metaheuristic framework?
4. What is the exploration vs. exploitation trade-off?
5. Why are metaheuristics effective for large-scale optimization problems?

Important Metaheuristics

Explain the following metaheuristics in detail:

1. Simulated Annealing
2. Genetic Algorithms
3. Tabu Search
4. Ant Colony Optimization
5. Particle Swarm Optimization
6. Greedy Randomized Adaptive Search Procedure (GRASP)
7. Variable Neighborhood Search
8. Iterated Local Search

For each method, describe:

- Main idea + parameters
- Include flow-chart and pseudocode
- Advantages and limitations
- Typical applications

4 Comparative Analysis

1. Compare:
 - Exact algorithms
 - Approximation algorithms
 - Heuristics
 - Metaheuristics
2. Discuss trade-offs in:
 - Solution quality
 - Computational time
 - Scalability
 - Theoretical guarantees
3. In which situations would you prefer:
 - Approximation algorithms?
 - Heuristics?
 - Metaheuristics?